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Semiconductor die having on-die power switch for selecting target operation voltage from operation voltages provided by different power sources

Abstract

A semiconductor die includes an on-die power switch and a target device. The on-die power switch includes a plurality of power input nodes, a power output node, and a switch circuit. The power input nodes receive a plurality of operation voltages from a plurality of different power sources, respectively. The power output node outputs a target operation voltage selected from the operation voltages. The switch circuit selectively couples one of the power input nodes to the power output node. The target device operates according to the target operation voltage supplied from the on-die power switch. The on-die power switch and the target device are separate circuit blocks of the semiconductor die.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8625368	12/2013	Campbell	N/A	N/A
9001572	12/2014	Choi	N/A	N/A
9171610	12/2014	Choi	N/A	N/A
9214221	12/2014	Yamaoka	N/A	N/A
9368194	12/2015	Komatsu	N/A	N/A
9570118	12/2016	Lee	N/A	N/A
10056907	12/2017	Asnaashari	N/A	N/A
2007/0262438	12/2006	Choa-Eoan	257/691	H03K 17/223
2009/0052246	12/2008	Bertin	N/A	N/A
2009/0085684	12/2008	Guo	331/176	H03L 1/02
2011/0111705	12/2010	Chua-Eoan	N/A	N/A
2014/0219010	12/2013	Yamaoka	365/154	G11C 11/417
2014/0313819	12/2013	Choi	365/156	G11C 5/14
2015/0035446	12/2014	Yamahara	N/A	N/A
2015/0036446	12/2014	Kenkare	N/A	N/A
2016/0179122	12/2015	Trivedi	N/A	N/A
2019/0220571	12/2018	Chung	N/A	N/A
2023/0050338	12/2022	Kong	N/A	H03K 3/35625
2023/0131586	12/2022	Pedersen	713/320	G06F 1/3296

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
6815544	12/2020	JP	N/A
201939299	12/2018	TW	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/347,047, filed on May 31, 2022. Further, this application claims the benefit of U.S. Provisional Application No. 63/432,040, filed on Dec. 12, 2022. The contents of these applications are incorporated herein by reference.

BACKGROUND

(1) The present invention relates to a semiconductor device, and more particularly, to a semiconductor die having an on-die power switch for selecting a target operation voltage from operation voltages provided by different power sources.

(2) When the technology node (a.k.a. the process node) goes smaller due to advance of the semiconductor process, the operation voltage of most devices goes lower during the low performance requirement. Regarding the low performance requirement, there are some devices that still need a minimum operation voltage that is higher than the operation voltage of other devices. To lower the power consumption, a conventional solution may adopt extra power sources to meet the requirement of multiple operation voltages for any device that needs an operation voltage varying with the performance requirement and still needs a minimum operation voltage during the low performance requirement, which leads to high power source cost, high printed circuit board (PCB) layout complexity, more PCB layers, and larger PCB area.

SUMMARY

(3) One of the objectives of the claimed invention is to provide a semiconductor die having an on-die power switch for selecting a target operation voltage from operation voltages provided by different power sources.

(4) According to a first aspect of the present invention, an exemplary semiconductor die is disclosed. The exemplary semiconductor die includes an on-die power switch and a target device. The on-die power switch includes a plurality of power input nodes, a power output node, and a switch circuit. The power input nodes are arranged to receive a plurality of operation voltages from a plurality of different power sources, respectively. The power output node is arranged to output a target operation voltage selected from the operation voltages. The switch circuit is arranged to selectively couple one of the power input nodes to the power output node. The target device is arranged to operate according to the target operation voltage supplied from the on-die power switch. The on-die power switch and the target device are separate circuit blocks of the semiconductor die.

(5) According to a second aspect of the present invention, an exemplary semiconductor die is disclosed. The exemplary semiconductor die includes a first device, an on-die power switch, and a target device. The first device is arranged to operate according to a first operation voltage from a first power source. The on-die power switch includes a plurality of power input nodes, a power output node, and a switch circuit. The power input nodes are arranged to receive a plurality of operation voltages from a plurality of different power sources, respectively, wherein the operation voltages include the first operation voltage. The power output node is arranged to output a target operation voltage selected from the operation voltages. The switch circuit is arranged to selectively couple one of the power input nodes to the power output node. The target device is arranged to operate according to the target operation voltage supplied from the on-die power switch.

(6) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram illustrating a first semiconductor die with an on-die power switch according to an embodiment of the present invention.

(2) FIG. **2** is a section view of a semiconductor die with an on-die power switch according to an embodiment of the present invention.

(3) FIG. **3** is a diagram illustrating different operation voltages V.sub.MIN, V.sub.DYN for different performance requirements according to an embodiment of the present invention.

(4) FIG. **4** is a diagram illustrating a second semiconductor die with an on-die power switch according to an embodiment of the present invention.

(5) FIG. **5** is a diagram illustrating different operation voltages V.sub.AO, V.sub.LG for different performance requirements according to an embodiment of the present invention.

(6) FIG. **6** is a diagram illustrating a third semiconductor die with an on-die power switch according to an embodiment of the present invention.

DETAILED DESCRIPTION

(7) Certain terms are used throughout the following description and claims, which refer to particular components. As one skilled in the art will appreciate, electronic equipment manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not in function. In the following description and in the claims, the terms “include” and “comprise” are used in an open-ended fashion, and thus should be interpreted to mean “include, but not limited to . . . ”. Also, the term “couple” is intended to mean either an indirect or direct electrical connection. Accordingly, if one device is coupled to another device, that connection may be through a direct electrical connection, or through an indirect electrical connection via other devices and connections.

(8) FIG. **1** is a diagram illustrating a first semiconductor die with an on-die power switch according to an embodiment of the present invention. The semiconductor die **100** may be a system on a chip (SoC) including a plurality of circuit designs integrated therein, where the circuit design may include one or more devices that require multiple operation voltages under different performance requirements. In this embodiment, the semiconductor die **100** includes a plurality of devices **102_1-102**_N (N≥2), **104** and a power switch **106**. It should be noted that only the components pertinent to the present invention are shown in FIG. **1**. In practice, the semiconductor die **100** may include additional components for other designated functions. The semiconductor die **100** is packaged in a semiconductor package **10**. For example, the semiconductor package **10** may be a flip-chip package. The devices **102_1-102**_N (N≥2), **104** and the power switch **106** are separate circuit blocks of the semiconductor die **100**. That is, the power switch **106** employs a centralized design, and is not integrated within any of the devices **102_1-102**_N (N≥2), **104**. The power switch **106** is an on-die power switch, and includes a plurality of power input nodes (e.g., power input pads) **108_1-108**_N (N≥2), a power output node (e.g., a power output pad) **110**, and a switch circuit **112**. The power input nodes **108_1-108**_N are arranged to receive a plurality of operation voltages V_**1**-V_N (N≥2) from a plurality of different power sources **12_1-12**_N (N≥2), respectively. Examples of the power sources **12_1-12**_N may include a buck converter, a low-dropout regulator, etc. The power output node **110** is arranged to output a target operation voltage V_IN selected from the operation voltages V_**1**-V_N available at the power input nodes **108_1-108**_N. The switch circuit **112** is arranged to selectively couple one of the power input nodes **108_1-108**_N to the power output node **110**. The device **104** may need different operation voltages under different performance requirements. For example, the device **104** may be a memory device such as a static random access memory (SRAM). When the device **104** is under a high performance requirement, the device **104** may operate according to a dynamic operation voltage (e.g., V_**1**) that varies with the performance requirement change. When the device **104** operates under a low performance requirement, the device **104** may operate according to a fixed operation voltage (e.g., V_N) that

does not vary with the performance requirement change, where the fixed operation voltage (e.g., V_N) may be a minimum operation voltage needed by the device **104** under the low performance requirement, and may be higher than the dynamic operation voltage (e.g., V_**1**) used by other devices under the low performance requirement.
(9) With the help of the on-die power switch **106**, the operation voltage V_**1** (e.g., dynamic operation voltage) used by the device **102_1** can be shared with the device **104** that can operate according to the operation voltage V_**1** under the high performance mode, and the operation voltage V_N (e.g., fixed operation voltage) used by the device **102**_N can be shared with the device **104** that needs to operate according to the operation voltage V_N under the low performance mode. In this way, the number of external power sources needed by the semiconductor die (e.g., SoC) **100** can be minimized, the PCB layout complexity can be reduced, and the power performance can be maintained or enhanced. Furthermore, since the power switch **106** and the devices **102_1-102**_N, **104** in the same semiconductor die **100** are fabricated by the same semiconductor process, there is no extra layout resource needed, and the power integrity (PI) simulation can be guaranteed.
(10) A wafer bumping process may be applied to a wafer before the wafer is being diced into individual semiconductor dies. For example, bumping is essential to the flip-chip semiconductor packaging. In a case where the semiconductor package **10** is a flip-chip package, bumps are formed on one surface of the semiconductor die **100**. In some embodiments of the present invention, each of the power input nodes **108_1-108**_N is a power input pad with one bump formed thereon, and the power output node **110** is a power output pad with a bump formed thereon. Compared to an embedded design of a power switch with no bump out, the proposed power switch design with bump out allows easy post-silicon verification.
(11) FIG. **2** is a section view of a semiconductor die with an on-die power switch according to an embodiment of the present invention. For better comprehension of technical features of the present invention, it is assumed that the power switch **106** may be implemented by an on-die dual power switch (labeled by “DPSW”) **206**; the device **104** may be implemented by two devices **204_1** and **204_2**, each being designed to operate according to a fixed operation voltage V.sub.MIN under a low performance requirement and operate according to a dynamic operation voltage V.sub.DYN under a high performance requirement, as illustrated in FIG. **3**; and the devices **102_1-102**_N may be implemented by devices **202_1** and **202_2**. A semiconductor die (labeled by “DIE”) is mounted on a PCB of a semiconductor package (labeled by “PKG”). Functions of the devices **202_1**, **202_2**, **204_1**, **204_2** and the on-die dual-power switch **206** in the same semiconductor die are realized by circuit layouts on metal layers. In addition, bumps are formed on pads that can be properly distributed on the surface of the semiconductor die through a redistribution layer (RDL). As shown in FIG. **2**, the on-die dual power switch **206** has one bump **208_1** formed on an input power pad **212_1** (which acts as one input power node of the power switch), one bump **208_2** formed on an input power pad **212_2** (which acts as another input power node of the power switch), and one bump **214** formed on an output power pad **214** (which acts as an output power node of the power switch). With a proper control of the on-die dual power switch **206** in the proximity of the target device (e.g., device **204_1**/**204_2**), different power requirements of the target device (e.g., device **204_1**/**204_2**) under different performance conditions can be met, without the use of additional power sources dedicated to the target device (e.g., device **204_1**/**204_2**).
(12) Furthermore, though the on-die dual power switch **206** is added to the semiconductor die, there is no need to modify the existing PCB layout of the semiconductor package.
(13) Designs for retaining data under a system standby/suspend state or working for an extremely light load application will usually be placed in an always-on power domain with an operation voltage supplied from an always-on power source. The always-on power source is usually a single and unique power source in a system to achieve minimum quiescent current of regulators. In addition, the always-on power source is generally not able to do dynamic voltage scaling (DVS)

since multiple circuit designs with multiple characteristics are placed in this always-on power domain. In certain cases, a circuit design may use a typical retention flip-flop which can hold its internal state when a primary power source is shut down and can have the ability to restore the state when the primary power source is brought up. The typical retention flip-flop, however, is generally designed to have two power rails, where one power rail is used to deliver an operation voltage supplied from the primary power source to a master flip-flop circuit in the retention flip-flop, and the other power rail is used to deliver an operation voltage supplied from an always-on power source to a shadow flip-flop circuit (i.e., a slave flip-flop circuit) in the same retention flip-flop. Since the master flip-flop circuit and the shadow flip-flop circuit are placed in distinct power domains, the typical retention flip-flop generally requires a level-shifter design between the master flip-flop circuit and the shadow flip-flop circuit and a power isolation design between two power rails, which results in a large die area inevitably. To address these issues, the aforementioned on-die power switch can be employed for allowing a circuit design to use a single-rail retention flip-flop with a smaller die area for retaining data under a system standby/suspend state and to have a function logic that can operate according to either a dynamic operation voltage when a non-always-on power source is turned on or an always-on operation voltage when the non-always-on power source is turned off. That is, with the help of the proposed on-die power switch, a circuit that employs DVS to save power can operate in an always-on power domain, and can employ a single-rail retention flip-flop to achieve reduction of the occupied die area. Further details are described as below with reference to the accompanying drawings.

(14) FIG. **4** is a diagram illustrating a second semiconductor die with an on-die power switch according to an embodiment of the present invention. The semiconductor die **400** may be an SoC including a plurality of circuit designs integrated therein, where the circuit design may include one or more devices that require multiple operation voltages under different operation states such as a normal state or a power saving state (e.g., standby/suspend state). In this embodiment, the semiconductor die **400** includes a plurality of devices **402_1**, **402_2**, **404** and a power switch **406**. It should be noted that only the components pertinent to the present invention are shown in FIG. **4**. In practice, the semiconductor die **400** may include additional components for other designated functions.

(15) The devices **402_1** and **402_2** act as an SRAM module, where the device **402_1** is an SRAM control circuit (labeled by “SRAM CTRL”), and the device **402_2** includes SRAM bit cells. The device **404** is arranged to perform a designated function, and includes a function logic **414**, a single-rail retention flip-flop (labeled by “SR-RTFF”) **416**, and a power electronic circuit **418**. The single-rail retention flip-flop **416** includes a power rail **420**, a master flip-flop circuit (labeled by “M-FF”) **422** and a shadow flip-flop circuit (labeled by “S-FF”) **424**, where the power rail **420** is arranged to deliver a target operation voltage V_IN selected and output from the power switch **406**, and is accessible to both of the master flip-flop circuit **422** and the shadow flip-flop circuit **424**. That is, supply voltages of both of the master flip-flop circuit **422** and the shadow flip-flop circuit **424** are obtained from the same power rail **420**. The power electronic circuit **418** may be implemented by MOS-controlled thyristors (MCTs), and can be used to control if the operation voltage V_IN is supplied to the function logic **414** and the master flip-flop circuit **422**.

(16) The semiconductor die **400** is packaged in a semiconductor package **40**. For example, the semiconductor package **40** may be a flip-chip package. The devices **402_1**, **402_2** (i.e., SRAM control circuit and SRAM bit cells) and the power switch **406** are separate circuit blocks of the semiconductor die **400**. That is, the power switch **406** employs a centralized design, and is not integrated within any of the devices **402_1**, **402_2**, **404**. The power switch **406** is an on-die power switch, and includes a plurality of power input nodes (e.g., power input pads) **408_1**, **408_2**, a power output node (e.g., a power output pad) **410**, and a switch circuit **412**. The power input nodes **408_1**, **408_2** are arranged to receive a plurality of operation voltages V.sub.LG and V.sub.AO from a plurality of different power sources **42_1**, **42_2**, respectively. Examples of the power

sources **42_1**, **42_2** may include a buck converter, a low-dropout regulator, etc. The power output node **410** is arranged to output the target operation voltage V_IN selected from the operation voltages V.sub.LG, V.sub.AO available at the power input nodes **408_1**, **408_2**. The switch circuit **412** is arranged to selectively couple one of the power input nodes **408_1**, **408_2** to the power output node **410**. For example, the power switch **406** may be implemented by the aforementioned power switch **106** or **206**. Since a person skilled in the art can readily understand details of the power switch **406** after reading above paragraphs directed to the power switch **106**/**206**, further description is omitted here for brevity.
(17) In this embodiment, the operation voltage V.sub.LG may be a dynamic operation voltage provided from the power source **42_1** that supports DVS, and the operation voltage V.sub.AO may be an always-on voltage that is a fixed operation voltage provided from the power source **42_2**. As shown in FIG. **4**, the operation voltage V.sub.AO is provided to the device **402_2** (i.e., SRAM bit cells) of the SRAM module **401** and the power input node **408_2** of the power switch **406**, and the operation voltage V.sub.LG is provided to the power input node **408_1** of the power switch **406**.
(18) When the device **404** operates under a normal mode, the power switch **406** (particularly, switch circuit **412** of power switch **406**) selects the operation voltage V.sub.LG (which is a dynamic operation voltage) as the target operation voltage V_IN. Regarding the device **404**, the operation voltage V.sub.LG is supplied to the function logic **414** and the master flip-flip circuit **422** under control of the power electronic circuit **418**, and is further supplied to the shadow flip-flop circuit **424**. It should be noted that, in this embodiment, both of the device (e.g., SRAM control circuit) **402_1** and the device **404** operate according to the same target operation voltage V_IN selected and output from the power switch **406**.
(19) When the device **404** operates under a power saving mode, the power switch **406** (particularly, switch circuit **412** of power switch **406**) selects the operation voltage V.sub.AO (which is a fixed operation voltage) as the target operation voltage V_IN. The power source **42_1** may be turned off to save quiescent current. Regarding the device **404**, the operation voltage V.sub.LG is not supplied to the function logic **414** and the master flip-flip circuit **422** under control of the power electronic circuit **418**, and is supplied to the shadow flip-flop circuit **424**. In this way, an internal state of the single-rail retention flip-flop **416** (particularly, an internal state of master flip-flip circuit **422**) is held by the shadow flip-flop circuit **424** that is powered by the operation voltage V.sub.AO. The same intended purpose of the typical dual-rail retention flip-flop can be achieved by using the proposed single-rail retention flip-flop **416**. Compared to the typical dual-rail retention flip-flop, the proposed single-rail retention flip-flop **416** does not require a level-shifter design between the master flip-flop circuit **422** and the shadow flip-flop circuit **424** due to the fact that both of the master flip-flop circuit **422** and the shadow flip-flop circuit **424** drain required power from the same power rail **420**, and does not require a power isolation design between two power rails due to the fact that there is only a single power rail **420**. Hence, the proposed single-rail retention flip-flop **416** occupies a much smaller die area compared to the typical dual-rail retention flip-flop.
(20) It should be noted that, when the power switch **406** (particularly, switch circuit **412** of power switch **406**) selects the operation voltage V.sub.AO (which is a fixed operation voltage) as the target operation voltage V_IN, the target operation voltage V_IN is delivered via the power rail **420** of the single-rail retention flip-flop **416**, and is accessible to the master flip-flop circuit **422** through the power electronic circuit **418**. In other words, when the power electronic circuit **418** turns on a path between the power rail **420** and the master flip-flop circuit **422**, the master flip-flop circuit **422** is allowed to operate in the always-on power domain. Similarly, when the power electronic circuit **418** turns on a path between the target operation voltage V_IN and the function logic **414**, the function logic **414** is allowed operate in the always-on power domain.
(21) With the help of the power switch **406** as well as the power electronic circuit **418**, the device **404** can operate according to the operation voltage V.sub.LA (which is a dynamic operation voltage that is supplied to the function logic **414** and the master flip-flop circuit **422** through the power

electronic circuit **418** and is further supplied to the shadow flip-flop circuit **424**) under system normal operation, as illustrated by the voltage range VR**1** and the performance range PR**1** shown in FIG. **5**; the device **404** can retain data according to the operation voltage V.sub.AO (which is an always-on voltage that is not supplied to the function logic **414** and the master flip-flop circuit **422** through the power electronic circuit **418** and is supplied to the shadow flip-flop circuit **424**) under system standby operation, as illustrated by the performance index P**0** shown in FIG. **5**; and the device **404** can normally operate according to the operation voltage V.sub.AO (which is an always-on voltage that is supplied to the function logic **414** and the master flip-flop circuit **422** through the power electronic circuit **418** and is further supplied to the shadow flip-flop circuit **424**), as illustrated by the performance range PR**2** shown in FIG. **5**.

(22) Regarding the embodiment shown in FIG. **4**, the device **402_2** (i.e., SRAM bit cells) is placed in the always-on power domain, and operates according to the operation voltage V.sub.AO only. However, this is for illustrative purposes only, and is not meant to be a limitation of the present invention.

(23) FIG. **6** is a diagram illustrating a third semiconductor die with an on-die power switch according to an embodiment of the present invention. The semiconductor die **600** may be an SoC including a plurality of circuit designs integrated therein, where the circuit design may include one or more devices that require multiple operation voltages under different operation states such as a normal state or a power saving state (e.g., standby/suspend state). The major difference between the semiconductor die **600** packaged in the semiconductor package **60** and the semiconductor die **400** packaged in the semiconductor package **40** is that the semiconductor die **600** further includes a power switch **606**. It should be noted that only the components pertinent to the present invention are shown in FIG. **6**. In practice, the semiconductor die **600** may include additional components for other designated functions.

(24) The devices **402_1**, **402_2** (i.e., SRAM control circuit and SRAM bit cells) and the power switches **406**, **606** are separate circuit blocks of the semiconductor die **600**. That is, each of the power switches **406**, **606** employs a centralized design, and is not integrated within any of the devices **402_1**, **402_2**, **404**. Like the power switch **406**, the power switch **606** is an on-die power switch, including a plurality of power input nodes (e.g., power input pads) **608_1**, **608_2**, a power output node (e.g., a power output pad) **610**, and a switch circuit **612**. The power input nodes **608_1**, **608_2** are arranged to receive operation voltages V.sub.LG and V.sub.AO from power sources **42_1**, **42_2**, respectively. The power output node **610** is arranged to output another target operation voltage V_IN**2** selected from the operation voltages V.sub.LG, V.sub.AO available at the power input nodes **608_1**, **608_2**, where the target operation voltage V_IN**2** is supplied to the device **402_2** (i.e., SRAM: bit cells) of the SRAM module **401**. The switch circuit **612** is arranged to selectively couple one of the power input nodes **608_1**, **608_2** to the power output node **610**. For example, the power switch **606** may be implemented by the aforementioned power switch **106** or **206**. Since a person skilled in the art can readily understand details of the power switch **606** after reading above paragraphs directed to the power switch **106**/**206**, further description is omitted here for brevity.

(25) As mentioned above, the operation voltage V.sub.LG may be a dynamic operation voltage provided from the power source **42_1** that supports DVS, and the operation voltage V.sub.AO may be an always-on voltage that is a fixed operation voltage provided from the power source **42_2**. In this embodiment, the device **402_2** (i.e., SRAM bit cells) can operate according to a fixed operation voltage (i.e., operation voltage V.sub.AO being an always-on voltage) under a low performance requirement, as illustrated by the performance range PR**1** shown in FIG. **5**; and can operate according to a dynamic operation voltage (i.e., operation voltage V.sub.LG) under a high performance requirement, as illustrated by the voltage range VR**2** and the performance range PR**2** in FIG. **5**. In some embodiments of the present invention, when both of the power switches **406** and **606** select the operation voltage V.sub.AO as the target operation voltages V_IN and VIN_IN**2**, the

power source **42_1** may be turned off to save quiescent current. It should be noted that, in this embodiment, both of the device (e.g., SRAM control circuit) **402_1** and the device **404** operate according to the same target operation voltage V_IN selected and output from the power switch **406**, and/or the target operation voltage V_IN selected and output from the power switch **406** may be the same or different from the target operation voltage V_IN**2** selected and output from the power switch **606**, depending upon actual application requirements.
(26) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor die comprising: an on-die power switch, comprising: a plurality of power input nodes, arranged to receive a plurality of operation voltages from a plurality of different power sources, respectively, wherein the plurality of power input nodes are a plurality of power input pads, respectively; a power output node, arranged to output a target operation voltage selected from the plurality of operation voltages, wherein the power output node is a power output pad; and a switch circuit, arranged to selectively couple one of the plurality of power input nodes to the power output node; a target device, arranged to operate according to the target operation voltage supplied from the on-die power switch; and a plurality of bumps, wherein one bump is formed on each of the plurality of power input pads and the power output pad; wherein the on-die power switch and the target device are separate circuit blocks of the semiconductor die.
2. The semiconductor die of claim 1, wherein the target device is a memory device.
3. The semiconductor die of claim 2, wherein the memory device is a static random access memory (SRAM).
4. The semiconductor die of claim 1, wherein one of the plurality of operation voltages is a fixed operation voltage.
5. The semiconductor die of claim 1, wherein one of the plurality of operation voltages is a dynamic operation voltage.
6. The semiconductor die of claim 1, wherein the on-die power switch has only two power input nodes.
7. A semiconductor die comprising: an on-die power switch, comprising: a plurality of power input nodes, arranged to receive a plurality of operation voltages from a plurality of different power sources, respectively; a power output node, arranged to output a target operation voltage selected from the plurality of operation voltages; and a switch circuit, arranged to selectively couple one of the plurality of power input nodes to the power output node; and a target device, arranged to operate according to the target operation voltage supplied from the on-die power switch, wherein the on-die power switch and the target device are separate circuit blocks of the semiconductor die, one of the plurality of operation voltages is an always-on voltage, and the target device comprises: a function logic; a single-rail retention flip-flop, comprising: a master flip-flop circuit; a shadow flip-flop circuit; and a power rail, arranged to deliver the target operation voltage supplied from the on-die power switch, wherein the power rail is accessible to both of the master flip-flop circuit and the shadow flip-flop circuit; and a power electronic circuit, arranged to control if the target operation voltage is supplied to the function logic and the master flip-flop circuit; wherein when the target operation voltage is the always-on voltage, the target operation voltage is not supplied to the function logic and the master flip-flop circuit through the power electronic circuit and is supplied to the shadow flip-flop circuit; or the target operation voltage is supplied to the function logic and the master flip-flop circuit through the power electronic circuit, and is further supplied to the shadow flip-flop circuit.

8. The semiconductor die of claim 7, wherein when the target operation voltage is not the always-on voltage, the target operation voltage is supplied to the function logic and the master flip-flop circuit through the power electronic circuit, and is further supplied to the shadow flip-flop circuit.
9. A semiconductor die comprising: a first device, arranged to operate according to a first operation voltage from a first power source; an on-die power switch, comprising: a plurality of power input nodes, arranged to receive a plurality of operation voltages from a plurality of different power sources, respectively, wherein the plurality of operation voltages comprise the first operation voltage; a power output node, arranged to output a target operation voltage selected from the plurality of operation voltages; and a switch circuit, arranged to selectively couple one of the plurality of power input nodes to the power output node; a target device, arranged to operate according to the target operation voltage supplied from the on-die power switch, wherein the target device comprises: a function logic; a single-rail retention flip-flop, comprising: a master flip-flop circuit; a shadow flip-flop circuit; and a power rail, arranged to deliver the target operation voltage supplied from the on-die power switch, wherein the power rail is accessible to both of the master flip-flop circuit and the shadow flip-flop circuit; and a power electronic circuit, arranged to control if the target operation voltage is supplied to the function logic and the master flip-flop circuit; and another on-die power switch, comprising: a plurality of another power input nodes, arranged to receive the plurality of operation voltages comprising the first operation voltage; another power output node, arranged to output another target operation voltage selected from the plurality of operation voltages; and another switch circuit, arranged to selectively couple one of the plurality of another power input nodes to the another power output node; wherein the first device is further arranged to operate according to the another target operation voltage supplied from the another on-die power switch.
10. The semiconductor die of claim 9, wherein the on-die power switch has only two power input nodes.
11. The semiconductor die of claim 10, further comprising: a second device, arranged to operate according to a second operation voltage from a second power source; wherein the plurality of operation voltages further comprise the second operation voltage.
12. The semiconductor die of claim 9, wherein the target device is a memory device.
13. The semiconductor die of claim 12, wherein the memory device is a static random access memory (SRAM).
14. The semiconductor die of claim 9, wherein the first operation voltage is a fixed operation voltage.
15. The semiconductor die of claim 9, wherein the first operation voltage is a dynamic operation voltage.
16. A semiconductor die comprising: a first device, arranged to operate according to a first operation voltage from a first power source; an on-die power switch, comprising: a plurality of power input nodes, arranged to receive a plurality of operation voltages from a plurality of different power sources, respectively, wherein the plurality of operation voltages comprise the first operation voltage; a power output node, arranged to output a target operation voltage selected from the plurality of operation voltages; and a switch circuit, arranged to selectively couple one of the plurality of power input nodes to the power output node; and a target device, arranged to operate according to the target operation voltage supplied from the on-die power switch, wherein the target device comprises: a function logic; a single-rail retention flip-flop, comprising: a master flip-flop circuit; a shadow flip-flop circuit; and a power rail, arranged to deliver the target operation voltage supplied from the on-die power switch, wherein the power rail is accessible to both of the master flip-flop circuit and the shadow flip-flop circuit; and a power electronic circuit, arranged to control if the target operation voltage is supplied to the function logic and the master flip-flop circuit; wherein when the target device operates under a power saving mode due to selection of the target operation voltage, the target operation voltage is not supplied to the function logic and the master

flip-flop circuit through the power electronic circuit, and is supplied to the shadow flip-flop circuit; or the target operation voltage is supplied to the function logic and the master flip-flop circuit through the power electronic circuit, and is further supplied to the shadow flip-flop circuit.
17. The semiconductor die of claim 16, wherein the first device is a memory device.
18. The semiconductor die of claim 17, wherein the memory device is a static random access memory (SRAM).
19. The semiconductor die of claim 16, wherein the first device is further arranged to operate according to the target operation voltage supplied from the on-die power switch.
20. The semiconductor die of claim 16, further comprising: another on-die power switch, comprising: a plurality of another power input nodes, arranged to receive the plurality of operation voltages comprising the first operation voltage; another power output node, arranged to output another target operation voltage selected from the plurality of operation voltages; and another switch circuit, arranged to selectively couple one of the plurality of another power input nodes to the another power output node; wherein the first device is further arranged to operate according to the another target operation voltage supplied from the another on-die power switch.
21. The semiconductor die of claim 16, wherein when the target device operates under a normal mode due to selection of the target operation voltage, the target operation voltage is supplied to the function logic and the master flip-flop circuit through the power electronic circuit, and is further supplied to the shadow flip-flop circuit.